

Proof Gaussian integrates to $\sqrt{2\pi}$.

Easy demonstration that shows the reason for the $1/\sqrt{2\pi}$ normalization in the gaussian formula.

$$I(r) = \int_{-\infty}^{\infty} e^{\left(-\frac{1}{2r^2}(y-\mu)^2\right)} dy \quad (\text{Unnormalized gaussian})$$

$$\sim \frac{y-\mu}{r} = z, \quad y = rz + \mu, \quad \frac{dz}{dy} = \frac{d\left(\frac{y-\mu}{r}\right)}{dy} = \frac{1}{r} \sim dy = r \cdot dz$$

$$I(r) = \int_{-\infty}^{\infty} e^{\left(-\frac{z^2}{2}\right)} r \cdot dz = r \int_{-\infty}^{\infty} e^{\left(-\frac{z^2}{2}\right)} \cdot dz = r I_0$$

Squaring and changing variable name on I_0 :

$$I_0^2 = \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} dz \int_{-\infty}^{\infty} e^{-\frac{y^2}{2}} dz = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)/2} dx dy$$

Change of variable: multivariate case (Section 2.8.3.2).

$$p_y(\bar{y}) = p_x(\bar{g}(\bar{y})) \cdot \left| \det \left\{ \bar{J}_{\bar{g}}(\bar{y}) \right\} \right|, \quad \bar{J}_{\bar{g}}(\bar{y}) = \frac{\partial \bar{g}(\bar{y})}{\partial \bar{y}^T}$$

$$\sim x = r \cos \theta, \quad y = r \sin \theta$$

$$dx dy = \left| \det \left\{ J(x, y) \right\} \right| dr d\theta, \quad J = \begin{bmatrix} \frac{\partial x}{\partial r}, \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r}, \frac{\partial y}{\partial \theta} \end{bmatrix} = \begin{bmatrix} \cos \theta, -r \sin \theta \\ \sin \theta, r \cos \theta \end{bmatrix}$$

$$\det \{ J(x, y) \} = r \cos^2 \theta + r \sin^2 \theta = r \underbrace{(\cos^2 \theta + \sin^2 \theta)}_1 = r$$

$$dx dy = r dr d\theta$$

$$I_o^2 = \int_0^{2\pi} \int_0^{\infty} e^{-(r^2 \cos^2(\theta) + r^2 \sin^2(\theta))/2} r dr d\theta = \int_0^{2\pi} \int_0^{\infty} e^{-\frac{r^2}{2}} r dr d\theta = 2\pi \int_0^{\infty} e^{-\frac{r^2}{2}} r dr$$

$$\leadsto u = r^2/2, \quad \frac{du}{dr} = \frac{d(\frac{r^2}{2})}{dr} = r, \quad dr = \frac{1}{r} du$$

$$= 2\pi \int_0^{2\pi} e^{-u} du = 2\pi \cdot 1$$

$$I_o^2 = 2\pi \Rightarrow I_o = \sqrt{2\pi}$$

Going back to I :

$$\boxed{I = \sigma \sqrt{2\pi} = \sqrt{2\pi} \sigma^2}$$